ICPC Notebook

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dijkstra-负权.hpp

md5: dcdf40

```cpp
#include <bits/stdc++.h>
#define N 60006
#define ll long long
using namespace std;
const ll inf = 1e9;
struct edge {
   int v, next, w;
} e[N << 1];
struct node {
   ll pos, d;
   bool operator<(const node& x) const { return x.d < d; }
};
int head[N], cnt, n, m, tot[N];
ll h[N], dis[N];
bool vis[N];
void add(int u, int v, int w) {
   e[++cnt].v = v;
   e[cnt].w = w;
   e[cnt].next = head[u];
   head[u] = cnt;
}
bool spfa() {
   queue<int> q;
   memset(h, 0x7f, sizeof(h));
   q.push(0);
   h[0] = 0;
   vis[0] = 1;
   while(!q.empty()) {
      int u = q.front();
      q.pop();
      vis[u] = 0;
      for(int i = head[u]; i; i = e[i].next) {
         int v = e[i].v, w = e[i].w;
         if(h[v] > h[u] + w) {
            h[v] = h[u] + w;
            tot[v] = tot[u] + 1;
            if(tot[v] > n) return 1;
            if(vis[v] == 0) q.push(v), vis[v] = 1;
         }
      }
   }
   return 0;
}
void dijkstra(int s) {
   priority_queue<node> q;
   memset(vis, 0, sizeof(vis));
   for(int i = 1; i <= n; i++) dis[i] = inf;
   dis[s] = 0;
   q.push({s, 0});
   while(!q.empty()) {
      int u = q.top().pos;
      q.pop();
      if(vis[u] == 1) continue;
      vis[u] = 1;
      for(int i = head[u]; i; i = e[i].next) {
         int v = e[i].v, w = e[i].w;
         if(dis[v] > dis[u] + w) {
            dis[v] = dis[u] + w;
            if(vis[v] == 0) q.push({v, dis[v]});
         }
      }
   }
}
int main() {
```

```
   ios::sync_with_stdio(false);
   cin.tie(0);
   cout.tie(0);
   cin >> n >> m;
   for(int i = 1; i <= m; i++) {
      int x, y, z;
      cin >> x >> y >> z;
      add(x, y, z);
   }
   for(int i = 1; i <= n; i++) add(0, i, 0);
   if(spfa() == 1) {
      cout << -1;
      return 0;
   }
   for(int u = 1; u <= n; u++)
      for(int i = head[u]; i; i = e[i].next) e[i].w += h[u] -
h[e[i].v];
   for(int i = 1; i <= n; i++) {
      dijkstra(i);
      ll ans = 0;
      for(int j = 1; j <= n; j++) {
         if(dis[j] == inf) ans += j * inf;
         else ans += j * (dis[j] + h[j] - h[i]);
      }
      cout << ans << "\n";
   }
   return 0;
}
```

dijkstra.hpp

md5: dd41b7

```
#include <bits/stdc++.h>
#define N 500005
#define ll long long
using namespace std;
const ll inf = 2147483647;
struct edge {
   int v, next, w;
} e[N << 1];
struct node {
   int dis, pos;
   bool operator<(const node& x) const { return x.dis < dis; }
};
int n, m, head[N], cnt, s, dis[N];
bool vis[N];
priority_queue<node> q;
void add(int u, int v, int w) {
   e[++cnt].v = v;
   e[cnt].w = w;
   e[cnt].next = head[u];
   head[u] = cnt;
}
void dijkstra() {
   dis[s] = 0;
   q.push({0, s});
   while(!q.empty()) {
      int x = q.top().pos;
      q.pop();
      if(vis[x] == 1) continue;
      vis[x] = 1;
      for(int i = head[x]; i; i = e[i].next) {
         int v = e[i].v;
         if(dis[v] > dis[x] + e[i].w) {
            dis[v] = dis[x] + e[i].w;
            if(vis[v] == 0) q.push({dis[v], v});
         }
      }
   }
}
int main() {
   ios::sync_with_stdio(false);
   cin.tie(0);
   cout.tie(0);
   cin >> n >> m >> s;
   for(int i = 1; i <= n; i++) dis[i] = inf;
   for(int i = 1; i <= m; i++) {
      int x, y, z;
      cin >> x >> y >> z;
      add(x, y, z);
   }
   dijkstra();
   for(int i = 1; i <= n; i++) cout << dis[i] << " ";
   return 0;
}
```

floyd.hpp

md5: 196fe9

```
#include <bits/stdc++.h>
#define N 100005
#define ll long long
using namespace std;
const ll inf = 2147483647;
ll n, m, f[105][105];
int main() {
   ios::sync_with_stdio(false);
   cin.tie(0);
   cout.tie(0);
   cin >> n >> m;
   for(int i = 1; i <= n; i++)
      for(int j = 1; j <= n; j++)
         if(i != j) f[i][j] = inf;
   for(int i = 1; i <= m; i++) {
      ll x, y, z;
      cin >> x >> y >> z;
      f[x][y] = min(f[x][y], z);
      f[y][x] = min(f[y][x], z);
   }
   for(int k = 1; k <= n; k++) {
      for(int i = 1; i <= n; i++) {
         if(i == k || f[i][k] == inf) continue;
         for(int j = 1; j <= n; j++) { f[i][j] = min(f[i][j],
f[i][k] + f[k][j]); }
      }
   }
   for(int i = 1; i <= n; i++) {
      for(int j = 1; j <= n; j++) cout << f[i][j] << " ";
      cout << "\n";
   }
   return 0;
}
```

kruscal.hpp

md5: 31a4b1

```
#include <bits/stdc++.h>
#define N 200005
#define ll long long
using namespace std;
const ll inf = 2147483647;
struct edge {
   int u, v, w;
} e[N << 1];
int n, m, f[N], ans, cnt;
int find(int x) {
   if(f[x] == x) return x;
   return f[x] = find(f[x]);
}
bool cmp(edge a, edge b) { return a.w < b.w; }
void kruscal() {
   for(int i = 1; i <= m; i++) {
      int fu = find(e[i].u), fv = find(e[i].v);
      if(fu == fv) continue;
      f[fu] = fv;
      ans += e[i].w;
      if(++cnt == n - 1) break;
   }
}
int main() {
   ios::sync_with_stdio(false);
   cin.tie(0);
   cout.tie(0);
   cin >> n >> m;
   for(int i = 1; i <= n; i++) f[i] = i;
   for(int i = 1; i <= m; i++) cin >> e[i].u >> e[i].v >>
e[i].w;
   sort(e + 1, e + 1 + m, cmp);
   kruscal();
   if(cnt < n - 1) cout << "orz";
   else cout << ans;
   return 0;
}
```

prim.hpp

md5: e76df0

```cpp
#include <bits/stdc++.h>
#define N 200005
#define ll long long
using namespace std;
const ll inf = 2147483647;
struct edge {
   int v, next, w;
   bool operator<(const edge& x) const { return x.w < w; }
} e[N << 1];
int head[N], cnt, n, m, tot, now = 1, ans, dis[N];
bool vis[N];
void add(int u, int v, int w) {
   e[++cnt].v = v;
   e[cnt].w = w;
   e[cnt].next = head[u];
   head[u] = cnt;
}
void prim() {
   for(int i = 2; i <= n; i++) dis[i] = inf;
   for(int i = head[1]; i; i = e[i].next) dis[e[i].v] =
min(dis[e[i].v], e[i].w);
   while(++tot < n) {
      int minn = inf;
      vis[now] = 1;
      for(int i = 1; i <= n; i++) {
         if(vis[i] == 0 && minn > dis[i]) {
            minn = dis[i];
            now = i;
         }
      }
      if(minn == inf) break;
      ans += minn;
      for(int i = head[now]; i; i = e[i].next) {
         int v = e[i].v;
         if(dis[v] > e[i].w) dis[v] = e[i].w;
      }
   }
}
int main() {
   ios::sync_with_stdio(false);
   cin.tie(0);
   cout.tie(0);
   cin >> n >> m;
   for(int i = 1; i <= m; i++) {
      int x, y, z;
      cin >> x >> y >> z;
      add(x, y, z);
      add(y, x, z);
   }
   prim();
   if(tot == n - 1) cout << ans;
   else cout << "orz";
   return 0;
}
```

tarjan强连通分量.hpp

md5: 7ce3e4

```cpp
#include <bits/stdc++.h>
#define inf 2147483647
#define ll long long
#define N 100005
using namespace std;
const int mod = 998244353;
struct edge {
   int v, next;
} e[N << 1];
int n, m, cnt, head[N], num, s[N], len, dfn[N], tot, low[N],
scc[N];
vector<int> ans[N];
bool in[N], f[N];
void add(int u, int v) {
   e[++cnt].v = v;
   e[cnt].next = head[u];
   head[u] = cnt;
}
void dfs(int x) {
   dfn[x] = low[x] = ++tot;
   s[++len] = x;
   in[x] = 1;
   for(int i = head[x]; i; i = e[i].next) {
      int y = e[i].v;
      if(dfn[y] == 0) {
         dfs(y);
         low[x] = min(low[x], low[y]);
      } else if(in[y] == 1) low[x] = min(low[x], low[y]);
   }
   if(dfn[x] == low[x]) {
      ans[++num].push_back(x);
      while(x != s[len]) {
         scc[s[len]] = num;
         in[s[len]] = 0;
         ans[num].push_back(s[len]);
         len--;
      }
      len--;
      in[x] = 0;
      scc[x] = num;
   }
}
void solve() {
   cin >> n >> m;
   for(int i = 1; i <= m; i++) {
      int u, v;
      cin >> u >> v;
      add(u, v);
   }
   for(int i = 1; i <= n; i++)
      if(dfn[i] == 0) dfs(i);
   for(int i = 1; i <= n; i++) sort(ans[i].begin(),
ans[i].end());
   cout << num << "\n";
   for(int i = 1; i <= n; i++) {
      if(f[scc[i]] == 1) continue;
      f[scc[i]] = 1;
      for(int j = 0; j < ans[scc[i]].size(); j++) { cout <<
ans[scc[i]][j] << " "; }
      cout << "\n";
   }
}
signed main() {
   ios::sync_with_stdio(false);
   cin.tie(0);
   cout.tie(0);
   int T = 1;
   // cin>>T;
   while(T--) solve();
   return 0;
}
```

并查集.hpp

md5: 6b2234

```cpp
#include <bits/stdc++.h>
#define N 2000005
#define ll long long
using namespace std;
const ll inf = 2147483647;
int n, m, f[N];
int find(int x) {
   if(f[x] == x) return x;
   return f[x] = find(f[x]);
}
int main() {
   ios::sync_with_stdio(false);
   cin.tie(0);
   cout.tie(0);
   cin >> n >> m;
   for(int i = 1; i <= n; i++) f[i] = i;
   while(m--) {
      int z, x, y;
      cin >> z >> x >> y;
      if(z == 1) f[find(y)] = find(x);
      if(z == 2) {
         if(find(x) == find(y)) cout << "Y\n";
         else cout << "N\n";
      }
   }
   return 0;
}
```

拓扑排序.hpp

md5: 6c9ce8

```cpp
#include <bits/stdc++.h>
using namespace std;
queue<int> q;
int deg[1001], cnt, head[1001], n;
struct edge {
   int v, next, u;
} e[2002];
void add(int u, int v) {
   e[++cnt].v = v;
   e[cnt].next = head[u];
   head[u] = cnt;
}
void topo() {
   for(int i = 1; i <= n; i++) {
      if(deg[i] == 0) {
         q.push(i);
         cout << i << " ";
      }
   }
   while(!q.empty()) {
      int x = q.front();
      q.pop();
      for(int i = head[x]; i; i = e[i].next) {
         deg[e[i].v]--;
         if(deg[e[i].v] == 0) {
            q.push(e[i].v);
            cout << e[i].v << " ";
         }
      }
   }
}
int main() {
   ios::sync_with_stdio(false);
   cin.tie(0);
   cout.tie(0);
   cin >> n;
   for(int i = 1; i <= n; i++) {
      int x;
      while(cin >> x) {
         if(x == 0) break;
         deg[x]++;
         add(i, x);
      }
   }
   topo();
   return 0;
}
```

树上启发式合并.hpp

md5: ae2321

```cpp
#include <bits/stdc++.h>
#define int long long
using namespace std;
const int mod = 1e9 + 7;
const int inf = 1e18;
const int N = 200005;
struct node {
   int v, next;
} e[N << 1];
int n, head[N], sz[N], tot, c[N], m, cnt[N], hson[N], ans[N],
ccnt;
void add(int u, int v) {
   e[++tot] = {v, head[u]};
   head[u] = tot;
}
void dfs1(int u, int fa) {
   sz[u] = 1;
   for(int i = head[u]; i; i = e[i].next) {
      int v = e[i].v;
      if(v == fa) continue;
      dfs1(v, u);
      sz[u] += sz[v];
      if(sz[v] > sz[hson[u]]) hson[u] = v;
   }
}
void upd(int u, int fa, int x) {
   cnt[c[u]] += x;
   if(x == 1 && cnt[c[u]] == 1) ccnt++;
   if(x == -1 && cnt[c[u]] == 0) ccnt--;
   for(int i = head[u]; i; i = e[i].next) {
      int v = e[i].v;
      if(v == fa) continue;
      upd(v, u, x);
   }
}
void dfs2(int u, int fa, int pd) {
   for(int i = head[u]; i; i = e[i].next) {
      int v = e[i].v;
      if(v == fa || v == hson[u]) continue;
      dfs2(v, u, 0);
   }
   if(hson[u]) dfs2(hson[u], u, 1);
   for(int i = head[u]; i; i = e[i].next) {
      int v = e[i].v;
      if(v == fa || v == hson[u]) continue;
      upd(v, u, 1);
   }
   cnt[c[u]] += 1;
   if(cnt[c[u]] == 1) ccnt++;
   ans[u] = ccnt;
   if(pd == 0) upd(u, fa, -1);
}
void solve() {
   cin >> n;
   for(int i = 1; i < n; i++) {
      int x, y;
      cin >> x >> y;
      add(x, y), add(y, x);
   }
   for(int i = 1; i <= n; i++) cin >> c[i];
   dfs1(1, 0);
   dfs2(1, 0, 0);
   cin >> m;
   while(m--) {
      int u;
      cin >> u;
      cout << ans[u] << "\n";
   }
}
signed main() {
   ios::sync_with_stdio(false);
   cin.tie(0);
   cout.tie(0);
   int T = 1;
   // cin>>T;
   while(T--) solve();
   return 0;
}
```

缩点.hpp

md5: 3df608

```cpp
#include <bits/stdc++.h>
#define inf 2147483647
#define ll long long
#define N 100005
using namespace std;
const int mod = 998244353;
struct edge {
   int v, next;
} e[N << 1], newe[N << 1];
int n, m, head[N], newhead[N], cnt, newcnt, num, scc[N],
dfn[N], low[N], tot, val[N], s[N], len, dp[N], a[N];
bool in[N];
void add(int u, int v) {
   e[++cnt].v = v;
   e[cnt].next = head[u];
   head[u] = cnt;
}
void newadd(int u, int v) {
   newe[++newcnt].v = v;
   newe[newcnt].next = newhead[u];
   newhead[u] = newcnt;
}
void dfs(int x) {
   dfn[x] = low[x] = ++tot;
   s[++len] = x;
   in[x] = 1;
   for(int i = head[x]; i; i = e[i].next) {
      int y = e[i].v;
      if(dfn[y] == 0) {
         dfs(y);
```

```
            low[x] = min(low[x], low[y]);
        } else if(in[y] == 1) low[x] = min(low[x], low[y]);
    }
    if(dfn[x] == low[x]) {
        num++;
        while(s[len] != x) {
            in[s[len]] = 0;
            scc[s[len]] = num;
            val[num] += a[s[len]];
            len--;
        }
        scc[x] = num;
        val[num] += a[x];
        in[x] = 0;
        len--;
    }
}
void solve() {
    cin >> n >> m;
    for(int i = 1; i <= n; i++) cin >> a[i];
    for(int i = 1; i <= m; i++) {
        int u, v;
        cin >> u >> v;
        add(u, v);
    }
    for(int i = 1; i <= n; i++)
        if(dfn[i] == 0) dfs(i);
    set<pair<int, int>> st;
    for(int i = 1; i <= n; i++) {
        for(int j = head[i]; j; j = e[j].next) {
            if(scc[i] != scc[e[j].v]) {
                if(st.count({scc[i], scc[e[j].v]}) == 1) continue;
                st.insert({scc[i], scc[e[j].v]});
                newadd(scc[i], scc[e[j].v]);
            }
        }
    }
    for(int i = 1; i <= num; i++) dp[i] = val[i];
    for(int i = num; i >= 1; i--) {
        for(int j = newhead[i]; j; j = newe[j].next) {
dp[newe[j].v] = max(dp[newe[j].v], dp[i] + val[newe[j].v]); }
    }
    int ans = 0;
    for(int i = 1; i <= num; i++) ans = max(ans, dp[i]);
    cout << ans << "\n";
}
signed main() {
    ios::sync_with_stdio(false);
    cin.tie(0);
    cout.tie(0);
    int T = 1;
    // cin>>T;
    while(T--) solve();
    return 0;
}
```

虚树.hpp

md5: 13a69f

```
#include <bits/stdc++.h>
#define int long long
using namespace std;
const int mod = 1e9 + 7;
const int inf = 1e18;
const int N = 250005;
struct Tree {
    int cnt, head[N];
    struct edge {
        int v, w, next;
    } e[N << 1];
    void add(int u, int v, int w) {
        e[++cnt] = {v, w, head[u]};
        head[u] = cnt;
    }
} e1, e2;
int n, m, idx, top, st[N], a[N], dfn[N], fa[N][18], lg[N],
mn[N], dp[N], dep[N];
void dfs1(int u, int father) {
    dfn[u] = ++idx;
    fa[u][0] = father;
    dep[u] = dep[father] + 1;
    for(int i = 1; i <= lg[dep[u]]; i++) fa[u][i] = fa[fa[u][i -
1]][i - 1];
    for(int i = e1.head[u]; i; i = e1.e[i].next) {
        int v = e1.e[i].v;
        if(v == father) continue;
        mn[v] = min(mn[u], e1.e[i].w);
        dfs1(v, u);
    }
}
int lca(int x, int y) {
    if(dep[x] < dep[y]) swap(x, y);
    while(dep[x] > dep[y]) x = fa[x][lg[dep[x] - dep[y]]];
    if(x == y) return x;
    for(int i = lg[dep[x]]; i >= 0; i--)
        if(fa[x][i] != fa[y][i]) x = fa[x][i], y = fa[y][i];
    return fa[x][0];
}
bool cmp(int x, int y) { return dfn[x] < dfn[y]; }
void insert(int u) {
    if(top == 1) {
        if(u != 1) st[++top] = u;
        return;
    }
    int x = lca(u, st[top]);
    if(x == st[top]) return;
    for(; top > 1 && dfn[st[top - 1]] >= dfn[x]; top--)
e2.add(st[top - 1], st[top], 0);
    if(st[top] != x) e2.add(x, st[top], 0), st[top] = x;
    st[++top] = u;
}
void dfs2(int u) {
    if(!e2.head[u]) {
        dp[u] = mn[u];
        return;
    }
    dp[u] = 0;
    for(int i = e2.head[u]; i; i = e2.e[i].next) {
        int v = e2.e[i].v;
        dfs2(v);
        dp[u] += dp[v];
    }
    dp[u] = min(dp[u], mn[u]);
    e2.head[u] = 0;
}
void solve() {
    cin >> n;
    for(int i = 1; i < n; i++) {
        int u, v, w;
        cin >> u >> v >> w;
        e1.add(u, v, w), e1.add(v, u, w);
    }
    mn[1] = inf;
    dfs1(1, 0);
    cin >> m;
    while(m--) {
        int k;
        cin >> k;
        for(int i = 1; i <= k; i++) cin >> a[i];
        sort(a + 1, a + 1 + k, cmp);
        e2.cnt = 0;
        st[top = 1] = 1;
        for(int i = 1; i <= k; i++) insert(a[i]);
        for(; top > 1; top--) e2.add(st[top - 1], st[top], 0);
        dfs2(1);
        cout << dp[1] << "\n";
    }
}
signed main() {
    for(int i = 2; i <= N; i++) lg[i] = lg[i >> 1] + 1;
    ios::sync_with_stdio(false);
    cin.tie(0);
    cout.tie(0);
    int T = 1;
    // cin>>T;
    while(T--) solve();
    return 0;
}
```

负环-spfa.hpp

md5: ce6dbb

```
#include <bits/stdc++.h>
#define N 60006
#define ll long long
using namespace std;
```

```
const ll inf = 2147483647;
struct edge {
   int v, next, w;
} e[N << 1];
int head[N], cnt, tot[N], dis[N], n, m;
bool vis[N];
queue<int> q;
void add(int u, int v, int w) {
   e[++cnt].v = v;
   e[cnt].w = w;
   e[cnt].next = head[u];
   head[u] = cnt;
}
bool spfa() {
   vis[1] = 1;
   dis[1] = 0;
   q.push(1);
   while(!q.empty()) {
      int u = q.front();
      q.pop();
      vis[u] = 0;
      for(int i = head[u]; i; i = e[i].next) {
         int v = e[i].v;
         if(dis[v] > dis[u] + e[i].w) {
            dis[v] = dis[u] + e[i].w;
            tot[v] = tot[u] + 1;
            if(tot[v] >= n) return 1;
            if(vis[v] == 0) vis[v] = 1, q.push(v);
         }
      }
   }
   return 0;
}
void solve() {
   memset(tot, 0, sizeof(tot));
   memset(head, 0, sizeof(head));
   memset(vis, 0, sizeof(vis));
   memset(e, 0, sizeof(e));
   memset(dis, 0x3f, sizeof(dis));
   cnt = 0;
   cin >> n >> m;
   for(int i = 1; i <= m; i++) {
      int x, y, z;
      cin >> x >> y >> z;
      add(x, y, z);
      if(z >= 0) add(y, x, z);
   }
   if(spfa() == 1) cout << "YES\n";
   else cout << "NO\n";
}
int main() {
   ios::sync_with_stdio(false);
   cin.tie(0);
   cout.tie(0);
   int t;
   cin >> t;
   while(t--) solve();
   return 0;
}
```

graph/tree

LCA-dfn.hpp

md5: 78b45a

```
#include <bits/stdc++.h>
#define N 500005
using namespace std;
int lg[N], mi[22][N], n, m, s, dfn[N], dn, head[N], cnt;
struct edge {
   int v, next;
} e[N << 1];
int get(int x, int y) { return dfn[x] < dfn[y] ? x : y; }
void add(int u, int v) {
   e[++cnt].v = v;
   e[cnt].next = head[u];
   head[u] = cnt;
}
void dfs(int now, int father) {
   dfn[now] = ++dn;
   mi[0][dfn[now]] = father;
   for(int i = head[now]; i; i = e[i].next)
      if(e[i].v != father) dfs(e[i].v, now);
}
int lca(int x, int y) {
   if(x == y) return x;
   if(dfn[x] > dfn[y]) swap(x, y);
   int deep = lg[dfn[y] - dfn[x] - 1];
   return get(mi[deep][dfn[x] + 1], mi[deep][dfn[y] - (1 <<
deep) + 1]);
}
int main() {
   ios::sync_with_stdio(false);
   cin.tie(0);
   cout.tie(0);
   cin >> n >> m >> s;
   for(int i = 1; i < n; i++) {
      int x, y;
      cin >> x >> y;
      add(x, y);
      add(y, x);
   }
   for(int i = 2; i <= n; i++) lg[i] = lg[i >> 1] + 1;
   dfs(s, 0);
   for(int i = 1; i <= lg[n]; i++) {
      for(int j = 1; j <= n - (1 << i) + 1; j++) { mi[i][j] =
get(mi[i - 1][j], mi[i - 1][j + (1 << i - 1)]); }
   }
   for(int i = 1; i <= m; i++) {
      int x, y;
      cin >> x >> y;
      cout << lca(x, y) << "\n";
   }
   return 0;
}
```

LCA-tarjan.hpp

md5: 832c2e

```
#include <bits/stdc++.h>
#define MAXN 500005
using namespace std;
int n, m, s, cnt, vcnt, head[MAXN], vhead[MAXN], f[MAXN],
lca[MAXN << 2];
bool vis[MAXN];
struct edge {
   int to, next, vto, vnext;
} e[MAXN << 1];
void add(int u, int v) {
   e[++cnt].to = v;
   e[cnt].next = head[u];
   head[u] = cnt;
}
void vadd(int u, int v) {
   e[++vcnt].vto = v;
   e[vcnt].vnext = vhead[u];
   vhead[u] = vcnt;
}
int find(int x) {
   if(f[x] == x) return x;
   return f[x] = find(f[x]);
}
void tarjan(int u) {
   vis[u] = 1;
   for(int i = head[u]; i; i = e[i].next) {
      int v = e[i].to;
      if(vis[v] == 1) continue;
      tarjan(v);
      f[v] = u;
   }
   for(int i = vhead[u]; i; i = e[i].vnext) {
      int v = e[i].vto;
      if(vis[v] == 0) continue;
      lca[i] = find(v);
      if(i % 2 == 1) lca[i + 1] = lca[i];
      else lca[i - 1] = lca[i];
   }
}
int main() {
   ios::sync_with_stdio(false);
   cin.tie(0);
   cout.tie(0);
   cin >> n >> m >> s;
   for(int i = 1; i <= n; i++) f[i] = i;
   for(int i = 1; i < n; i++) {
```

```
        int x, y;
        cin >> x >> y;
        add(x, y);
        add(y, x);
    }
    for(int i = 1; i <= m; i++) {
        int x, y;
        cin >> x >> y;
        vadd(x, y);
        vadd(y, x);
    }
    tarjan(s);
    for(int i = 1; i <= m; i++) cout << lca[i * 2] << "\n";
    return 0;
}
```

LCA-倍增.hpp

md5: 7b6e10

```
#include <bits/stdc++.h>
#define MAXN 500005
using namespace std;
struct edge {
    int v, next;
} e[MAXN << 1];
int dep[MAXN], head[MAXN], f[MAXN][22], lg[MAXN], n, m, s, cnt;
void add(int u, int v) {
    e[++cnt].v = v;
    e[cnt].next = head[u];
    head[u] = cnt;
}
void dfs(int now, int father) {
    f[now][0] = father;
    dep[now] = dep[father] + 1;
    for(int i = 1; i <= lg[dep[now]]; i++) f[now][i] = f[f[now]
[i - 1]][i - 1];
    for(int i = head[now]; i; i = e[i].next) {
        if(e[i].v != father) dfs(e[i].v, now);
    }
}
int lca(int x, int y) {
    if(dep[x] < dep[y]) swap(x, y);
    while(dep[x] > dep[y]) x = f[x][lg[dep[x] - dep[y]]];
    if(x == y) return x;
    for(int i = lg[dep[x]]; i >= 0; i--) {
        if(f[x][i] != f[y][i]) {
            x = f[x][i];
            y = f[y][i];
        }
    }
    return f[x][0];
}
int main() {
    ios::sync_with_stdio(false);
    cin.tie(0);
    cout.tie(0);
    cin >> n >> m >> s;
    for(int i = 1; i < n; i++) {
        int x, y;
        cin >> x >> y;
        add(x, y);
        add(y, x);
    }
    for(int i = 2; i <= n; i++) lg[i] = lg[i >> 1] + 1;
    dfs(s, 0);
    for(int i = 1; i <= m; i++) {
        int x, y;
        cin >> x >> y;
        cout << lca(x, y) << "\n";
    }
    return 0;
}
```

树链剖分.hpp

md5: 475d70

```
#include <bits/stdc++.h>
#define N 200005
#define ll long long
using namespace std;
const ll inf = 2147483647;
struct edge {
    int v, next;
} e[N << 1];
int n, m, root, mod, ans[N << 2], tag[N << 2], son[N], id[N],
fa[N], cnt, dep[N], siz[N], top[N], head[N], w[N], wt[N];
int ls(int p) { return p << 1; }
int rs(int p) { return p << 1 | 1; }
void push_up(int p) {
    ans[p] = ans[ls(p)] + ans[rs(p)];
    ans[p] %= mod;
}
void f(int l, int r, int p, int k) {
    ans[p] += (r - l + 1) * k;
    ans[p] %= mod;
    tag[p] += k;
}
void push_down(int l, int r, int p) {
    int mid = (l + r) >> 1;
    f(l, mid, ls(p), tag[p]);
    f(mid + 1, r, rs(p), tag[p]);
    tag[p] = 0;
}
void update(int l, int r, int p, int L, int R, int k) {
    if(L <= l && r <= R) {
        f(l, r, p, k);
        return;
    }
    int mid = (l + r) >> 1;
    push_down(l, r, p);
    if(L <= mid) update(l, mid, ls(p), L, R, k);
    if(R > mid) update(mid + 1, r, rs(p), L, R, k);
    push_up(p);
}
int query(int l, int r, int p, int L, int R) {
    if(L <= l && r <= R) return ans[p];
    int res = 0, mid = (l + r) >> 1;
    push_down(l, r, p);
    if(L <= mid) res += query(l, mid, ls(p), L, R), res %= mod;
    if(R > mid) res += query(mid + 1, r, rs(p), L, R), res %=
mod;
    return res % mod;
}
void build(int l, int r, int p) {
    if(l == r) {
        ans[p] = wt[l] % mod;
        return;
    }
    int mid = (l + r) >> 1;
    build(l, mid, ls(p));
    build(mid + 1, r, rs(p));
    push_up(p);
}
void add(int u, int v) {
    e[++cnt].v = v;
    e[cnt].next = head[u];
    head[u] = cnt;
}
void dfs1(int u, int father) {
    dep[u] = dep[father] + 1;
    fa[u] = father;
    siz[u] = 1;
    int hson = -1;
    for(int i = head[u]; i; i = e[i].next) {
        int v = e[i].v;
        if(v == father) continue;
        dfs1(v, u);
        siz[u] += siz[v];
        if(siz[v] > hson) hson = siz[v], son[u] = v;
    }
}
void dfs2(int u, int topc) {
    id[u] = ++cnt;
    wt[cnt] = w[u];
    top[u] = topc;
    if(son[u] == 0) return;
    dfs2(son[u], topc);
    for(int i = head[u]; i; i = e[i].next) {
        int v = e[i].v;
        if(v == fa[u] || v == son[u]) continue;
        dfs2(v, v);
    }
}
void update1(int x, int y, int k) {
    k %= mod;
```

```
   while(top[x] != top[y]) {
      if(dep[top[x]] < dep[top[y]]) swap(x, y);
      update(1, n, 1, id[top[x]], id[x], k);
      x = fa[top[x]];
   }
   if(dep[x] > dep[y]) swap(x, y);
   update(1, n, 1, id[x], id[y], k);
}
int query1(int x, int y) {
   int res = 0;
   while(top[x] != top[y]) {
      if(dep[top[x]] < dep[top[y]]) swap(x, y);
      res += query(1, n, 1, id[top[x]], id[x]);
      res %= mod;
      x = fa[top[x]];
   }
   if(dep[x] > dep[y]) swap(x, y);
   res += query(1, n, 1, id[x], id[y]);
   return res % mod;
}
void update2(int x, int k) { update(1, n, 1, id[x], id[x] +
siz[x] - 1, k); }
int query2(int x) { return query(1, n, 1, id[x], id[x] + siz[x]
- 1); }
int main() {
   ios::sync_with_stdio(false);
   cin.tie(0);
   cout.tie(0);
   cin >> n >> m >> root >> mod;
   for(int i = 1; i <= n; i++) cin >> w[i];
   for(int i = 1; i < n; i++) {
      int x, y;
      cin >> x >> y;
      add(x, y);
      add(y, x);
   }
   cnt = 0;
   dfs1(root, 0);
   dfs2(root, root);
   build(1, n, 1);
   while(m--) {
      int op, x, y, z;
      cin >> op;
      if(op == 1) {
         cin >> x >> y >> z;
         update1(x, y, z);
      }
      if(op == 2) {
         cin >> x >> y;
         cout << query1(x, y) << "\n";
      }
      if(op == 3) {
         cin >> x >> y;
         update2(x, y);
      }
      if(op == 4) {
         cin >> x;
         cout << query2(x) << "\n";
      }
   }
   return 0;
}
```

flow

string

algorithm

geometry

memo